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Database Management Systems

AIRLINE RESERVATION SYSTEM

ER Diagram Design and Relational Schema

(DBMS Assignment 2)

***Bachelor of Computer Application
Semester – III***

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1. Objective

This assignment builds on the domain selected in DBMS Assignment 1 — the Airline Reservation System — by designing a complete Entity Relationship (ER) Diagram for the system and converting it into a formal Relational Schema. It identifies all entities, attributes, primary keys, relationships and cardinalities, and expresses the design as relational tables with primary and foreign keys, including the SQL DDL needed to create them.

2. Entities, Attributes and Primary Keys

The Airline Reservation System is composed of nine entities. Each entity's attributes, data types, and keys are listed below (PK = Primary Key, FK = Foreign Key).

1. Passenger

Attribute	Data Type	Key	Description
Passenger_ID	INT	PK	Unique passenger ID
First_Name	VARCHAR(50)	—	Passenger's first name
Last_Name	VARCHAR(50)	—	Passenger's last name
Email	VARCHAR(100)	—	Passenger's email
Phone	VARCHAR(15)	—	Passenger's phone number
Date_of_Birth	DATE	—	Passenger's date of birth
Gender	VARCHAR(10)	—	Passenger's gender

Primary Key: Passenger_ID

2. Airport

Attribute	Data Type	Key	Description
Airport_ID	INT	PK	Unique airport ID
Airport_Code	VARCHAR(10)	—	Airport code such as BLR
Airport_Name	VARCHAR(100)	—	Name of airport
City	VARCHAR(50)	—	City where airport is located
Country	VARCHAR(50)	—	Country
Terminal_Count	INT	—	Number of terminals

Primary Key: Airport_ID

3. Aircraft

Attribute	Data Type	Key	Description
Aircraft_ID	INT	PK	Unique aircraft ID
Aircraft_Model	VARCHAR(50)	—	Aircraft model
Registration_No	VARCHAR(30)	—	Aircraft registration number

Attribute	Data Type	Key	Description
Total_Seats	INT	—	Total seating capacity
Aircraft_Status	VARCHAR(20)	—	Active / Maintenance

Primary Key: Aircraft_ID

4. Flight

Attribute	Data Type	Key	Description
Flight_ID	INT	PK	Unique flight ID
Flight_Number	VARCHAR(20)	—	Airline flight number
Aircraft_ID	INT	FK	Aircraft assigned to the flight
Departure_Airport_ID	INT	FK	Departure airport
Arrival_Airport_ID	INT	FK	Arrival airport
Departure_DateTime	DATETIME	—	Departure date and time
Arrival_DateTime	DATETIME	—	Arrival date and time
Flight_Status	VARCHAR(20)	—	Scheduled / Delayed / Cancelled

Primary Key: Flight_ID

5. Seat

Attribute	Data Type	Key	Description
Seat_ID	INT	PK	Unique seat ID
Aircraft_ID	INT	FK	Aircraft containing the seat
Seat_Number	VARCHAR(10)	—	Seat number such as 12A
Seat_Class	VARCHAR(20)	—	Economy / Business / First
Seat_Status	VARCHAR(20)	—	Available / Reserved

Primary Key: Seat_ID

6. Booking (Associative Entity)

Attribute	Data Type	Key	Description
Booking_ID	INT	PK	Unique booking ID
Passenger_ID	INT	FK	Passenger making the booking
Flight_ID	INT	FK	Booked flight
Seat_ID	INT	FK	Selected seat
Booking_Date	DATE	—	Date booking was made
Booking_Status	VARCHAR(20)	—	Confirmed / Cancelled / Pending
Fare	DECIMAL(10,2)	—	Amount charged

Primary Key: Booking_ID

7. Ticket

Attribute	Data Type	Key	Description
Ticket_ID	INT	PK	Unique ticket ID
Booking_ID	INT	FK	Related booking
Ticket_Number	VARCHAR(30)	—	Ticket number
Issue_Date	DATE	—	Date ticket was issued
Ticket_Status	VARCHAR(20)	—	Valid / Cancelled

Primary Key: Ticket_ID

8. Payment

Attribute	Data Type	Key	Description
Payment_ID	INT	PK	Unique payment ID
Booking_ID	INT	FK	Related booking
Payment_Date	DATE	—	Date payment was made
Payment_Method	VARCHAR(30)	—	Card / UPI / Net Banking
Amount	DECIMAL(10,2)	—	Amount paid
Payment_Status	VARCHAR(20)	—	Successful / Failed / Pending
Transaction_ID	VARCHAR(100)	—	Payment transaction reference

Primary Key: Payment_ID

9. Baggage

Attribute	Data Type	Key	Description
Baggage_ID	INT	PK	Unique baggage ID
Booking_ID	INT	FK	Related booking
Baggage_Count	INT	—	Number of bags
Total_Weight	DECIMAL(5,2)	—	Total baggage weight
Excess_Baggage_Fee	DECIMAL(10,2)	—	Additional baggage charge

Primary Key: Baggage_ID

3. Relationships and Cardinalities

The table below lists every relationship between entities, its name, its cardinality, and a short explanation.

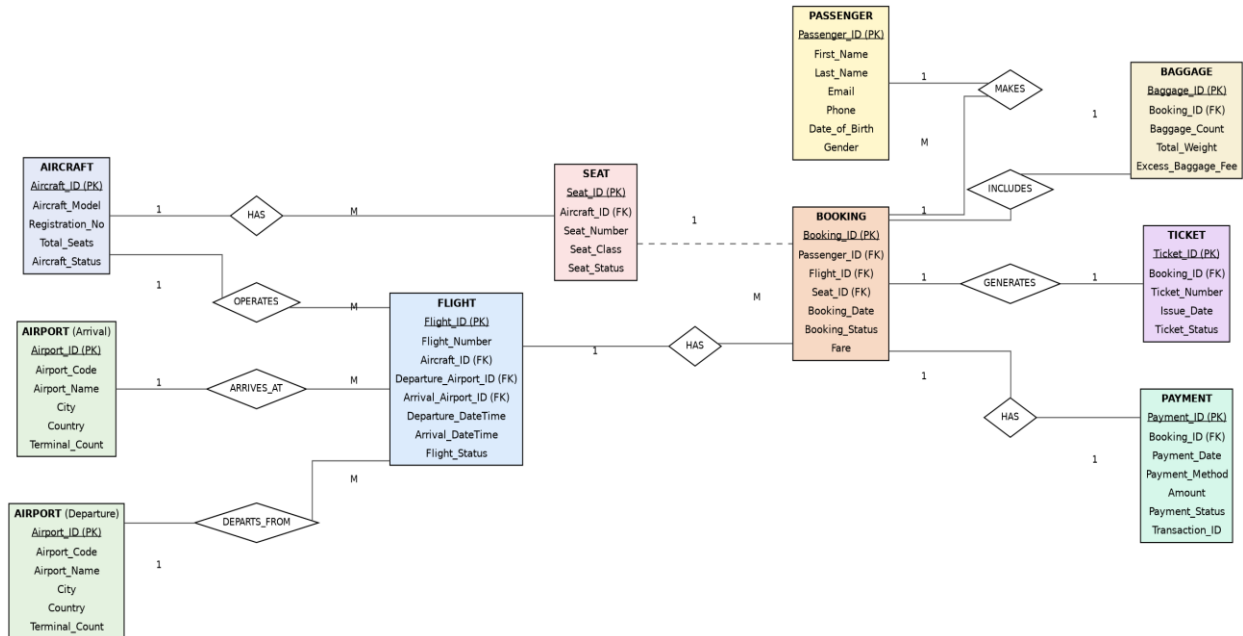
Entities	Relationship	Cardinality	Explanation
Passenger — Booking	MAKES	1 : M	One passenger can make many bookings.
Aircraft — Flight	OPERATES	1 : M	One aircraft operates many flights over time.
Aircraft — Seat	HAS	1 : M	One aircraft contains many seats.
Airport — Flight (Departure)	DEPARTS_FROM	1 : M	One airport is the departure point for many flights.
Airport — Flight (Arrival)	ARRIVES_AT	1 : M	One airport is the arrival point for many flights.
Flight — Booking	HAS	1 : M	One flight can have many bookings.
Seat — Booking	—	1 : M	One seat can be booked across different flights over time.
Booking — Ticket	GENERATES	1 : 1	A confirmed booking generates exactly one ticket.
Booking — Payment	HAS	1 : 1	A booking has one payment record.
Booking — Baggage	INCLUDES	1 : 1	A booking may have one associated baggage record.
Passenger — Flight	(via Booking)	M : N	Resolved by the Booking associative entity — a passenger books many flights, a flight carries many passengers.

Key DBMS concept:

Passenger and Flight share a many-to-many (M:N) relationship, since one passenger can book many flights and one flight carries many passengers. This cannot be represented directly in a relational database, so the Booking table is introduced as an associative (junction) entity: it carries its own primary key (Booking_ID) and holds the foreign keys Passenger_ID and Flight_ID (plus Seat_ID), converting the M:N relationship into two 1:M relationships — Passenger→Booking and Flight→Booking.

4. ER Diagram

The ER diagram below uses Chen notation: rectangles represent entities (with attributes listed inside and the primary key underlined), diamonds represent relationships, and the numbers on connecting lines show cardinality (1 or M). The dashed line between Seat and Booking reflects that a seat's involvement in a booking is tracked as a foreign key inside Booking rather than through a separate relationship diamond.



5. Relational Schema — Tables, Primary Keys and Foreign Keys

Each entity in the ER diagram maps to one relational table. The many-to-many relationship between Passenger and Flight is resolved through the Booking table, which acts as the associative table carrying both foreign keys.

Table	Primary Key (PK)	Foreign Key(s) (FK)
Passenger	Passenger_ID	—
Airport	Airport_ID	—
Aircraft	Aircraft_ID	—
Flight	Flight_ID	Aircraft_ID, Departure_Airport_ID, Arrival_Airport_ID
Seat	Seat_ID	Aircraft_ID
Booking	Booking_ID	Passenger_ID, Flight_ID, Seat_ID
Ticket	Ticket_ID	Booking_ID
Payment	Payment_ID	Booking_ID
Baggage	Baggage_ID	Booking_ID

6. SQL DDL Statements

The relational schema expressed as SQL CREATE TABLE statements, ready to run in MySQL Workbench or any relational database engine:

```
CREATE TABLE Passenger (  
    Passenger_ID      INT PRIMARY KEY,  
    First_Name        VARCHAR(50),  
    Last_Name         VARCHAR(50),  
    Email             VARCHAR(100),  
    Phone             VARCHAR(15),  
    Date_of_Birth     DATE,  
    Gender            VARCHAR(10)  
);  
  
CREATE TABLE Airport (  
    Airport_ID        INT PRIMARY KEY,  
    Airport_Code       VARCHAR(10),  
    Airport_Name       VARCHAR(100),  
    City              VARCHAR(50),  
    Country            VARCHAR(50),  
    Terminal_Count    INT  
);  
  
CREATE TABLE Aircraft (  
    Aircraft_ID       INT PRIMARY KEY,  
    Aircraft_Model     VARCHAR(50),  
    Registration_No    VARCHAR(30),  
    Total_Seats        INT,  
    Aircraft_Status    VARCHAR(20)  
);  
  
CREATE TABLE Flight (  
    Flight_ID          INT PRIMARY KEY,  
    Flight_Number       VARCHAR(20),  
    Aircraft_ID         INT,  
    Departure_Airport_ID INT,  
    Arrival_Airport_ID  INT,  
    Departure_DateTime  DATETIME,  
    Arrival_DateTime    DATETIME,  
    Flight_Status       VARCHAR(20),  
    FOREIGN KEY (Aircraft_ID) REFERENCES Aircraft(Aircraft_ID),  
    FOREIGN KEY (Departure_Airport_ID) REFERENCES Airport(Airport_ID),  
    FOREIGN KEY (Arrival_Airport_ID) REFERENCES Airport(Airport_ID)  
);  
  
CREATE TABLE Seat (  
    Seat_ID            INT PRIMARY KEY,  
    Aircraft_ID         INT,  
    Seat_Number         VARCHAR(10),  
    Seat_Class          VARCHAR(20),  
    Seat_Status         VARCHAR(20),  
    FOREIGN KEY (Aircraft_ID) REFERENCES Aircraft(Aircraft_ID)  
);  
  
CREATE TABLE Booking (  
    Booking_ID         INT PRIMARY KEY,  
    Passenger_ID        INT,  
    Flight_ID           INT,  
    Seat_ID             INT,  
    Booking_Date        DATE,  
    Booking_Status      VARCHAR(20),  
    Fare                DECIMAL(10,2),
```

```

FOREIGN KEY (Passenger_ID) REFERENCES Passenger(Passenger_ID),
FOREIGN KEY (Flight_ID) REFERENCES Flight(Flight_ID),
FOREIGN KEY (Seat_ID) REFERENCES Seat(Seat_ID)
);

CREATE TABLE Ticket (
    Ticket_ID      INT PRIMARY KEY,
    Booking_ID     INT,
    Ticket_Number  VARCHAR(30),
    Issue_Date     DATE,
    Ticket_Status  VARCHAR(20),
    FOREIGN KEY (Booking_ID) REFERENCES Booking(Booking_ID)
);

CREATE TABLE Payment (
    Payment_ID     INT PRIMARY KEY,
    Booking_ID     INT,
    Payment_Date   DATE,
    Payment_Method VARCHAR(30),
    Amount         DECIMAL(10,2),
    Payment_Status VARCHAR(20),
    Transaction_ID VARCHAR(100),
    FOREIGN KEY (Booking_ID) REFERENCES Booking(Booking_ID)
);

CREATE TABLE Baggage (
    Baggage_ID     INT PRIMARY KEY,
    Booking_ID     INT,
    Baggage_Count  INT,
    Total_Weight   DECIMAL(5,2),
    Excess_Baggage_Fee DECIMAL(10,2),
    FOREIGN KEY (Booking_ID) REFERENCES Booking(Booking_ID)
);

```

7. Relationship Explanation

- Passenger → Booking (1:M): One passenger can make many bookings over time, but each booking belongs to exactly one passenger.
- Aircraft → Flight (1:M): One aircraft can be assigned to operate many flights across its service life, but each flight uses exactly one aircraft.
- Aircraft → Seat (1:M): One aircraft has many seats, but each seat belongs to exactly one aircraft.
- Airport → Flight, twice (1:M each): One airport can be the departure point for many flights, and (in a separate relationship) the arrival point for many flights. Flight therefore carries two foreign keys referencing Airport: `Departure_Airport_ID` and `Arrival_Airport_ID`.
- Flight → Booking (1:M): One flight can be booked by many passengers (many bookings), but each booking is for exactly one flight.
- Seat → Booking (1:M): A given seat can appear in many bookings across different flights and dates, but within one booking only one seat is selected.
- Booking → Ticket (1:1): Once a booking is confirmed, exactly one ticket is generated for it.
- Booking → Payment (1:1): Each booking is settled by one payment record.
- Booking → Baggage (1:1): Each booking may have one baggage record describing the bags checked in for that trip.
- Passenger ↔ Flight (M:N, resolved): Directly, many passengers can travel on many flights. This is resolved by the Booking associative entity, which breaks the M:N relationship into two 1:M relationships as explained in Section 3.

8. Conclusion

The Airline Reservation System's ER diagram identifies nine entities — Passenger, Airport, Aircraft, Flight, Seat, Booking, Ticket, Payment, and Baggage — connected through a mix of one-to-many and one-to-one relationships, with the Booking table resolving the underlying many-to-many relationship between Passenger and Flight. Converting this diagram into a relational schema produces nine normalized tables in which primary keys uniquely identify each record and foreign keys enforce referential integrity between related tables. This design minimizes data duplication, keeps the data consistent, and supports efficient querying for flight search, reservations, payments, ticketing, and baggage handling.